

An iterated quadratic abc -orbit and a Lucas–Wieferich obstruction

Magnus Gille
Independent researcher

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Abstract

The elementary transfer

$$(a, b, a + b) \mapsto (4ab, (a - b)^2, (a + b)^2)$$

is iterated from $(1, 8, 9)$. Writing the resulting triples as (a_n, b_n, c_n) , we prove the exact radical formula

$$\frac{\text{rad}(a_n b_n c_n)}{c_n} = \frac{2}{3} \frac{\prod_{j < n} |T_{2^j}(7/9)|}{\prod_{j < n} |d_j| / \text{rad}(d_j)}, \quad d_j = a_j - b_j.$$

In particular every member of the orbit is an abc -hit. An explicit application of the Baker–Wüstholz theorem shows that the logarithm of the Chebyshev product is $O(n^2)$, whereas $\log c_n = 2^n \log 9$. Thus the asymptotic abc -quality of this orbit is governed, up to $O(n^2)$, only by repeated prime factors of the pairwise coprime integers d_j . These integers form the dyadic subsequence

$$|d_j| = \frac{1}{2} |V_{2^{j+1}}(2, 9)|$$

of a Lucas V -sequence. A prime divisor has prescribed dyadic order in $\mathbb{Q}(\sqrt{-2})$, and a square divisor is exactly a Wieferich-type failure of order growth. A reproducible search found no such square divisor for primes $p \leq 10^7$ and indices $0 \leq j \leq 50$. The computation is not used in the proofs.

1 Introduction

For a positive integer m , let $\text{rad}(m) = \prod_{p|m} p$. A coprime triple of positive integers (a, b, c) with $a + b = c$ is called an abc -hit when $\text{rad}(abc) < c$, and its quality is

$$q(a, b, c) = \frac{\log c}{\log \text{rad}(abc)}.$$

The abc conjecture predicts that, for every $\varepsilon > 0$, only finitely many such triples satisfy $c > \text{rad}(abc)^{1+\varepsilon}$; see, for example, Masser’s formulation [4].

The identity

$$4ab + (a - b)^2 = (a + b)^2 \tag{1.1}$$

transfers one additive triple to another. Polynomial transfers of this kind have long been used in searches for abc -triples; see van der Horst [6, §2.3]. Neither (1.1) nor the linear recurrence appearing below is new. In particular, the sequence $3^m T_m(1/3) = V_m(2, 9)/2$ is OEIS A025172 [5].

This note analyzes what happens when (1.1) is iterated from the small hit $(1, 8, 9)$. Its contribution is an exact separation of the radical loss into an archimedean Chebyshev factor and a

nonarchimedean repeated-prime factor. A quantitative linear-form estimate makes the first factor sub-power-sized. The remaining question is then a radical problem for a fixed Lucas sequence at dyadic indices. We also give an elementary order criterion which identifies square factors as number-field Lucas–Wieferich phenomena.

The conclusions are deliberately limited. We prove an infinite family of *abc*-hits, but no fixed quality gap. In particular, the results neither prove nor disprove the *abc* conjecture.

2 The quadratic orbit

Set

$$(a_0, b_0, c_0) = (1, 8, 9)$$

and, for $n \geq 0$, define

$$(a_{n+1}, b_{n+1}, c_{n+1}) = (4a_n b_n, (a_n - b_n)^2, c_n^2). \quad (2.1)$$

Identity (1.1) gives $a_n + b_n = c_n$ at every index. Put

$$d_n = a_n - b_n.$$

Then

$$c_n = 9^{2^n}, \quad d_{n+1} = c_n^2 - 2d_n^2. \quad (2.2)$$

Lemma 1. *Each d_n is odd and coprime to 6, and the integers d_n are pairwise coprime.*

Proof. The assertions about 2 and 3 follow from $d_0 = -7$ and (2.2): all c_n are odd and divisible by 3, while $d_{n+1} \equiv d_n^2 \equiv 1 \pmod{3}$.

Suppose that a prime p divides d_m . Since $p \nmid c_m$, (2.2) gives

$$d_{m+1} \equiv c_{m+1} \pmod{p}, \quad d_{m+2} \equiv -c_{m+2} \pmod{p}.$$

If $d_k \equiv -c_k \pmod{p}$, then (2.2) gives $d_{k+1} \equiv -c_{k+1} \pmod{p}$. Hence p divides no d_k with $k > m$. Interchanging two indices proves pairwise coprimality. $\square$

Lemma 2. *Every triple in (2.1) is primitive and*

$$\text{rad}(a_n b_n c_n) = 6 \prod_{j=0}^{n-1} \text{rad}(|d_j|), \quad (2.3)$$

where an empty product is 1.

Proof. For $n = 0$, the prime support of $a_0 b_0$ is $\{2\}$. For $n \geq 1$, $b_n = d_{n-1}^2$, while the recurrence for a_n in (2.1) shows inductively that the prime support of a_n is

$$\{2\} \cup \bigcup_{j=0}^{n-2} \{p : p \mid d_j\}.$$

Lemma 1 makes these supports disjoint and excludes 3. The only prime in c_n is 3, which proves both claims. $\square$

Normalize (2.2) by setting $x_n = -d_n/c_n$. Since $T_2(x) = 2x^2 - 1$,

$$x_{n+1} = T_2(x_n), \quad x_0 = \frac{7}{9},$$

and therefore

$$-\frac{d_n}{c_n} = T_{2^n} \left(\frac{7}{9} \right). \quad (2.4)$$

3 The exact radical identity

Define

$$t_j = \frac{|d_j|}{c_j} = \left| T_{2^j} \left(\frac{7}{9} \right) \right|, \quad Q_n = \prod_{j=0}^{n-1} \frac{|d_j|}{\text{rad}(|d_j|)}. \quad (3.1)$$

Thus Q_n records exactly the prime multiplicities beyond the first within $d_0, \dots, d_{n-1}$.

Theorem 3 (Exact separation). *For every $n \geq 0$,*

$$\boxed{\frac{\text{rad}(a_n b_n c_n)}{c_n} = \frac{2}{3} \frac{\prod_{j=0}^{n-1} t_j}{Q_n}}. \quad (3.2)$$

In particular, (a_n, b_n, c_n) is an abc-hit for every n .

Proof. As $\prod_{j < n} c_j = 9^{2^n - 1} = c_n/9$, (3.1) gives

$$\prod_{j < n} \text{rad}(|d_j|) = \frac{\prod_{j < n} |d_j|}{Q_n} = \frac{c_n}{9} \frac{\prod_{j < n} t_j}{Q_n}.$$

Substitution into (2.3) proves (3.2). For $n = 0$, its right side is $2/3$. For $n > 0$, each t_j lies strictly between 0 and 1; this also follows from the non-root-of-unity argument in the proof of Theorem 4. Since $Q_n \geq 1$, the ratio in (3.2) is strictly less than 1. $\square$

4 The archimedean factor

The estimate in this section is stated with an explicit, if very large, constant so that the dependence on the index is unambiguous.

Theorem 4 (Archimedean factor). *Let*

$$C_{\text{BW}} = 18 \cdot 3! \cdot 2^3 \cdot 64^4 \log 8, \quad \kappa = C_{\text{BW}} (\log 3) \frac{\pi}{2}.$$

Then, for $j \geq 0$,

$$t_j > \frac{1}{\pi} 2^{-\kappa(j+1)}. \quad (4.1)$$

Consequently,

$$0 < - \sum_{j=0}^{n-1} \log t_j < n \log \pi + \frac{\kappa \log 2}{2} n(n+1). \quad (4.2)$$

Proof. Put

$$z = \frac{7 + 4\sqrt{-2}}{9} = e^{i\theta}, \quad \theta = \arccos(7/9) \in (0, \pi/2).$$

Then $z + z^{-1} = 14/9$. The number z is not a root of unity: otherwise $z + z^{-1}$ would be both a rational number and an algebraic integer, hence an integer.

For an integer $N \geq 1$, choose an odd integer k nearest to $2N\theta/\pi$, and use the principal logarithms $\log z = i\theta$ and $\log(-1) = i\pi$. The linear form

$$\Lambda_N = 2N \log z - k \log(-1) \quad (4.3)$$

is nonzero, satisfies $|\Lambda_N| \leq \pi$, and has $|k| \leq N + 1 \leq 2N$.

We apply the explicit Baker–Wüstholz estimate [1, Theorem 1]. In its standard notation, the number field $\mathbb{Q}(z, -1) = \mathbb{Q}(\sqrt{-2})$ has degree $D = 2$, there are two logarithms, and the coefficient bound may be taken as $B = 2N$. The minimal polynomial $9X^2 - 14X + 9$ has both roots on the unit circle, so

$$h(z) = \log 3, \quad \max\{h(z), |\log z|/D, 1/D\} = \log 3.$$

For -1 , the corresponding modified height is $\pi/2$. The theorem therefore gives

$$\log |\Lambda_N| > -C_{\text{BW}}(\log 3) \frac{\pi}{2} \log(2N) = -\kappa \log(2N). \quad (4.4)$$

Because k is odd,

$$|z^{2N} + 1| = |1 - e^{\Lambda_N}|.$$

For a purely imaginary Λ with $|\Lambda| \leq \pi$, $|1 - e^{\Lambda}| \geq 2|\Lambda|/\pi$. Hence (4.4) yields

$$\frac{1}{2}|z^{2N} + 1| > \frac{1}{\pi}(2N)^{-\kappa}. \quad (4.5)$$

Taking $N = 2^j$ and using

$$t_j = \frac{1}{2}|z^{2N} + 1|$$

proves (4.1). Summing its logarithm proves the upper bound in (4.2). The lower bound follows from $0 < t_j < 1$; equality $t_j = 1$ would make z a root of unity. $\square$

Corollary 5 (Quality reduction). *Let*

$$R_n = \text{rad}(a_n b_n c_n), \quad L_n = \log c_n = 2^n \log 9, \quad q_n = \frac{\log c_n}{\log R_n}.$$

Then

$$\log \frac{c_n}{R_n} = \log Q_n + E_n, \quad (4.6)$$

where

$$\log \frac{3}{2} \leq E_n < \log \frac{3}{2} + n \log \pi + \frac{\kappa \log 2}{2} n(n+1). \quad (4.7)$$

In particular,

$$\log \frac{c_n}{R_n} = \log Q_n + O(n^2), \quad (4.8)$$

and

$$q_n \longrightarrow 1 \iff \log Q_n = o(\log c_n). \quad (4.9)$$

More generally,

$$\limsup_{n \rightarrow \infty} (q_n - 1) > 0 \iff \limsup_{n \rightarrow \infty} \frac{\log Q_n}{\log c_n} > 0. \quad (4.10)$$

Proof. Taking logarithms in (3.2) gives (4.6) with

$$E_n = \log(3/2) - \sum_{j < n} \log t_j.$$

Theorem 4 gives (4.7)–(4.8). Since $n^2 = o(L_n)$, (4.9) follows from

$$q_n - 1 = \frac{\log(c_n/R_n)}{\log R_n}.$$

The same identity and (4.8) show that either side of (4.10) is positive exactly when the other is. $\square$

Remark 6. For a fixed $\delta > 0$, the exact inequality

$$q_n \geq 1 + \delta \iff \log Q_n + E_n \geq \frac{\delta}{1 + \delta} \log c_n$$

is sometimes a more useful form of the reduction.

5 Lucas sequence and order lifting

Let $V_m(P, Q)$ denote the Lucas V -sequence with

$$V_0 = 2, \quad V_1 = P, \quad V_m = PV_{m-1} - QV_{m-2}.$$

For $P = 2, Q = 9$, put

$$A_m = \frac{1}{2}V_m(2, 9) = 3^m T_m(1/3).$$

Thus $A_0 = A_1 = 1$ and $A_m = 2A_{m-1} - 9A_{m-2}$. This is the sequence OEIS A025172 [5].

Proposition 7. *For every $j \geq 0$,*

$$|d_j| = |A_{2^{j+1}}| = \frac{1}{2}|V_{2^{j+1}}(2, 9)|. \quad (5.1)$$

Proof. The composition rule $T_r(T_s(x)) = T_{rs}(x)$ and $T_2(1/3) = -7/9$ give

$$A_{2^{j+1}} = 9^{2^j} T_{2^j}(-7/9).$$

Comparing with (2.4), and using $T_m(-x) = (-1)^m T_m(x)$, proves (5.1). $\square$

The next proposition locates every prime factor in a thin congruence class and makes the Wieferich description precise. Put

$$K = \mathbb{Q}(\sqrt{-2}), \quad \alpha = 1 + 2\sqrt{-2}, \quad \beta = 1 - 2\sqrt{-2}, \quad u = \alpha/\beta.$$

Then $V_m(2, 9) = \alpha^m + \beta^m$.

Proposition 8 (Dyadic order). *Let p be a prime divisor of d_j , put $m = 2^{j+1}$, and let $\mathfrak{p}$ be a prime of $\mathcal{O}_K$ above p . Then $p \nmid 6$ and*

$$\text{ord}_{(\mathcal{O}_K/\mathfrak{p})^\times}(u) = 2m = 2^{j+2}. \quad (5.2)$$

Consequently,

$$\begin{cases} p \equiv 1 \pmod{2^{j+2}}, & p \text{ splits in } K, \\ p \equiv -1 \pmod{2^{j+2}}, & p \text{ is inert in } K. \end{cases} \quad (5.3)$$

Moreover,

$$p^2 \mid d_j \iff u^m \equiv -1 \pmod{\mathfrak{p}^2} \text{ for every } \mathfrak{p} \mid p, \quad (5.4)$$

and in that case the order in (5.2) does not acquire the usual factor p on lifting from $\mathfrak{p}$ to $\mathfrak{p}^2$.

Proof. Lemma 1 gives $p \nmid 6$. Since β has norm 9, it is invertible modulo $\mathfrak{p}$. From (5.1), $\alpha^m + \beta^m \equiv 0 \pmod{\mathfrak{p}}$, and hence $u^m \equiv -1 \pmod{\mathfrak{p}}$. As m is a power of 2, the order of u is exactly $2m$.

If p splits, the residue field has p elements, so $2m \mid p - 1$. If p is inert, conjugation is the Frobenius on the residue field and $u^{p+1} = u\bar{u} = 1$, so $2m \mid p + 1$. This proves (5.3).

Because p is unramified and $2\beta^m$ is a unit at every $\mathfrak{p} \mid p$, (5.1) gives (5.4). The reduction of u has order prime to p ; its order modulo $\mathfrak{p}^2$ is therefore either $2m$ or $2mp$. In the first case u^m is the unique lift of the element -1 of order 2, so it is -1 modulo $\mathfrak{p}^2$. The converse is immediate. $\square$

Thus a prime with $p^2 \mid d_j$ is naturally a Lucas–Wieferich prime for the algebraic number u , which is a unit away from 3. Classical primitive-divisor theory for Lucas and Lehmer sequences addresses the occurrence of new prime divisors rather than their aggregate multiplicities; see, for example, Bilu–Hanrot–Voutier [2]. In this orbit, Lemma 1 already makes every prime divisor new relative to all other dyadic terms. What (4.9)–(4.10) require instead is control of the total repeated-prime quotient Q_n . Related squarefree primitive divisor results in arithmetic dynamics are conditional on Vojta’s conjecture over number fields [3], and in any case the existence of one squarefree prime per term would be much weaker than the proportional radical estimate needed here.

6 Computation

The accompanying program `check_square_lifts.py` works entirely modulo p^2 . For each prime p , it starts with $(c, d) = (9, -7)$ and iterates

$$(c, d) \mapsto (c^2, c^2 - 2d^2) \pmod{p^2}.$$

It reports p, j precisely when $d \equiv 0 \pmod{p^2}$. If $d \equiv 0 \pmod{p}$ but not modulo p^2 , the search for that prime may stop, by Lemma 1.

Proposition 9 (Finite verification). *For every prime $p \leq 10^7$, $p \neq 2, 3$, and every $0 \leq j \leq 50$, the modular recurrence gives*

$$p^2 \nmid d_j.$$

The search tests 664,577 primes and reports zero square lifts.

This is a finite computer verification, not an input to any preceding theorem. The script uses a deterministic sieve and integer modular arithmetic only. Its command line is

```
python3 check_square_lifts.py -prime-limit 10000000 -max-j 50.
```

For orientation, the first terms factor as follows.

j	d_j	$ d_j $ factored
0	−7	7
1	−17	17
2	5983	$31 \cdot 193$
3	−28545857	$2753 \cdot 10369$
4	223288285122943	$127 \cdot 19841 \cdot 88613249$

7 Concluding question

The orbit has an exact dichotomy. Its quality tends to 1 if and only if the accumulated repeated-prime quotient of the dyadic Lucas terms is sub-power-sized:

$$\prod_{j < n} \frac{|V_{2^{j+1}}(2, 9)|/2}{\text{rad}(|V_{2^{j+1}}(2, 9)|/2)} = c_n^{o(1)}.$$

A fixed quality gap occurs along a subsequence if and only if the same product has positive-power growth. The Baker–Wüstholz estimate rules out the archimedean Chebyshev values as a source of such a gap; the remaining obstruction is purely the aggregate of Lucas–Wieferich lifts. We do not know how to prove either alternative.

AI-use statement

Claude Fable 5 and OpenAI Codex were used extensively to generate candidate arguments, conduct literature searches, write and run the computation, draft the manuscript, and perform reciprocal adversarial review. The named author directed the work and is responsible for the final content.

References

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